Hypothesis Testing

CSE 451
Topics in Computer Science
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What is Hypothesis Testing?

- Hypothesis testing is a part of statistical analysis, where we test the assumptions made regarding a population parameter.
- It is generally used when we were to compare:
  - a single group with an external standard
  - two or more groups with each other

Aside: Parameter vs. Statistic

- In statistics, a **parameter** is a description of a population, while a **statistic** describes a small portion of a population (**sample**).

- For example, if you ask everyone in your **class** (population) about their average height, you receive a parameter, a true description about the population since everyone was asked.
- If you now want to guess the average height of people in your **grade** (population) using the information you have from your **class** (sample), this information turns into a statistic.

What is a hypothesis

- A hypothesis is often described as an "educated guess" about a specific parameter or population.
- Once it is defined, one can collect data to determine whether it provides enough evidence that the hypothesis is true.

Null Hypothesis

- Null hypothesis is a statistical theory that suggests there exists no statistical significance between the populations.
- It is denoted by H_0 and read as H-naught.
- Null hypotheses often include phrases such as "no effect," "no difference," or "no relationship." When written in mathematical terms, they always include an equality (usually =, but sometimes ≥ or ≤).

Null Hypothesis: examples

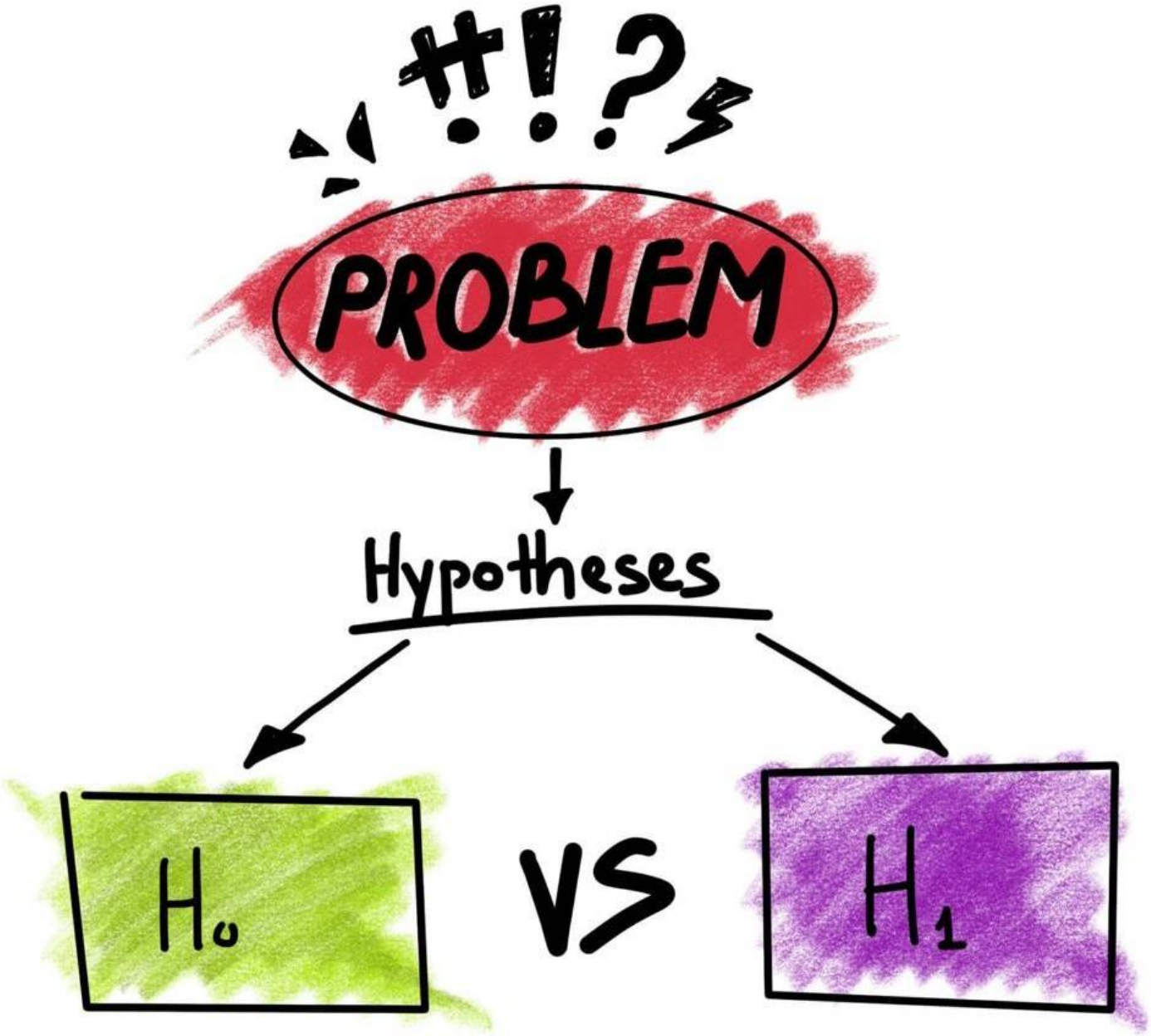
Research question	**Null hypothesis (H_0)**
Does tooth flossing affect the number of cavities?	Tooth flossing has **no effect** on the number of cavities.
Does the amount of text highlighted in the textbook affect exam scores?	The amount of text highlighted in the textbook has **no effect** on exam scores.
Does daily meditation decrease the incidence of depression?	Daily meditation **does not decrease** the incidence of depression.*

Alternative Hypothesis

- The alternative hypothesis (H_a or H_1) is the other answer to your research question. It claims that there's an effect in the population.
- Often, your alternative hypothesis is the same as your research hypothesis. In other words, it's the claim that you expect or hope will be true.
- The alternative hypothesis is the complement to the null hypothesis.
- Null and alternative hypotheses are mutually exclusive, meaning that only one can be true at a time.
- Alternative hypotheses often include phrases such as "an effect," "a difference," or "a relationship." When alternative hypotheses are written in mathematical terms, they always include an inequality (usually $\neq$, but sometimes $<$ or $>$)

Alternative Hypothesis: examples

Research question	**Alternative hypothesis (H_a)**
Does tooth flossing affect the number of cavities?	Tooth flossing has an **effect** on the number of cavities.
Does the amount of text highlighted in a textbook affect exam scores?	The amount of text highlighted in the textbook has an **effect** on exam scores.
Does daily meditation decrease the incidence of depression?	Daily meditation **decreases** the incidence of depression.


#!?
PROBLEM
Hypotheses
H_0
VS
H_1

Example: Number of College Applications

- To know how many colleges students applied to, the dean of a certain university took a random sample of size 106 from their newly admitted students.
- This sample yielded an average of 9.7 college applications with a standard deviation of 7.
- College Board website states that counselors recommend students apply to roughly 8 colleges.
- Do these data provide convincing evidence that the average number of colleges all freshmen in this university apply to is higher than recommended?

Example: Number of College Applications – Hypotheses

- Population: all freshmen in this university
- The parameter of interest μ is the average number of schools applied to by all freshmen in this university
- There are two explanations why the sample mean is higher than the recommended 8 schools.
  - The true population mean is different.
  - The true population mean is 8, and the difference between the true population mean and the sample mean is simply due to natural sampling variability.
- H_0 : $\mu = 8$ (the average number of colleges freshmen in this university have applied to is 8, as recommended)
- H_A : $\mu > 8$ (the average number of colleges freshmen in this university have applied to is > 8)

What to do after the formulation of the hypothesis?

Apply statistical tests to get the answer, i.e., hypothesis testing

Hypothesis Testing Again

- Given a pair of complementary hypothesis about a population, null (H_0) and alternative (H_1) hypotheses, hypothesis testing is the technique of analyzing sample data to make one of the following two decisions:
  - We have enough evidence to reject H_0 in favor of H_1.
  - We don't have enough evidence to reject H_0 in favor of H_1.

Case Study: Can Dogs Smell Bladder Cancer?

- A study by M. Willis et al. (British Medical Journal, vol. 329, September 25, 2004.) considered whether dogs could be trained to detect if a person has bladder cancer by smelling certain compounds in the patient's urine.
- 6 dogs of varying breeds were trained to discriminate between urine from patients with bladder cancer and urine from control patients without it.
- The dogs were taught to indicate which among several specimens was from the bladder cancer patient by lying beside it.
- Once trained, the dogs' ability to distinguish cancer patients from controls was tested using urine samples from subjects not previously encountered by the dogs.

Case Study: Can Dogs Smell Bladder Cancer?

- The researchers blinded both dog handlers and experimental observers to the identity of urine samples.
- Each of the 6 dogs was tested with 9 trials. In each trial, one urine sample from a bladder cancer patient was randomly placed among 6 control urine samples.
- Outcome: In the total of 54 trials with the 6 dogs, the dogs made the correct selection 22 times.
  - The dogs were correct for 22/54 ≈ 40.7% of the time.
  - If the dogs just guessed at random, they were expected to be correct for 1/7 ≈ 14.3% of the time
- Is this difference (40.7% v.s. 14.3%) surprising?

Two Competing Hypotheses

- Let p be the proportion of time that dogs were correct.

1. Null hypothesis (H_0): $p = 1/7$
   - "There is nothing going on."
   - The dogs just guessed at random.
   - "null" means "nothing surprising is going on".
   - The dogs were just lucky to make more correct selections than expected.
2. Alternative hypothesis (H_A): $p > 1/7$
   - "There is something going on."
   - Dogs can do better than random guessing.

Test Statistic

- The test statistic is a summary of the data that best reflect the evidence *for* or *against* the hypotheses.
- For this study, we use

  X = "the total number of correct selection in the 54 trials" as our test statistic.

- The larger X is, the stronger the evidence for H_A and against H_0
- The smaller X is, the stronger the evidence for H_0 and against H_A

p-values

- The probability $P(X \geq 22)$ is called the p-value of the test. What is a p-value?

- The p-value of the test is *the probability of observing data such that the evidence for the H_A is at least as strong as our current data set, assuming the H_0 is true.*
  - The p-value for the dog study is $P(X \geq 22)$ not $P(X = 22)$.

p-values

The smaller the p-value, the stronger the evidence against the H_0:

- A p-value of 0.25 says that if the H_0 was true, then we would obtain a sample that looks like the observed sample 1 in 4 of the time;

  $\Rightarrow$ the data seems to be consistent with H_0
- A p-value of 0.001 says that if the H_0 was true, then only 1 out of every 1,000 samples would resemble the observed sample;

  $\Rightarrow$ the H_0 looks doubtful

Significance Level

- So, the smaller the p-value is, the stronger the evidence against the null hypothesis.
- In some studies, we can simply report the p-value, and let people judge whether the evidence is strong enough
- In other studies, we need to make a decision about which hypothesis to trust
  - We then select a cut-off value α, call the significance level
  - If the p-value $< \alpha$, we reject H_0
  - If the p-value $> \alpha$, we don't reject H_0

Significance Level

- Commonly used significance levels: 0.05 and 0.01

- A test with p-value < 0.05 is said to be (statistically) significant
- A test with p-value < 0.01 is said to be highly significant

Type 1 and Type 2 Errors

- In a hypothesis test, we make a decision about which of H_0 or H_A might be true, but our decision might be incorrect.

		Decision	
		fail to reject H_0	reject H_0
Truth	H_0 true	✓	*Type 1 Error*
	H_A true	*Type 2 Error*	✓

- A Type 1 Error is rejecting the H_0 when it is true.
- A Type 2 Error is failing to reject the H_0 when it is false.
- We (almost) never know if H_0 or H_A is true, but we need to consider all possibilities

Consequences of Type 1 and 2 errors

- Type 1 and type 2 errors are different sorts of mistakes and have different consequences
- Usually H_0 is the status quo, a thing we generally believe to be true
- If H_0 is not rejected, usually it means the status quo is fine. No action needs to be taken
- Rejecting H_0 means something we use to believe is overturned. It might be a scientific breakthrough (e.g., discovery of a new drug).

Consequences of Type 1 and 2 errors

- A type 1 error introduces a false conclusion into the scientific community and can lead to a tremendous waste of resources before further research invalidates the original finding.
- A type 2 error — failing to recognize a scientific breakthrough — represents a missed opportunity for scientific progress

- Type 2 errors can be costly as well, but generally go unnoticed
- So it's more important to control the Type 1 error rate than the Type 2 error rate.

Significance Level = Type 1 Error Rate

- When using a 5% significance level, there is about 5% chance of making a Type 1 error if the H_0 is true.

$$P(\text{Type 1 error} \mid H_0 \text{ true}) = \alpha$$

- This is why we prefer small values of α — increasing α increases the Type 1 error rate.

- However, the significance level doesn't control Type 2 error rate

True or False

If H_0 is rejected, then we can be certain that H_0 is false.

- FALSE Even if H_0 is true, 5% of the time the experiment will give a result with a p-value < 5% so that H_0 is rejected.

Recap: Hypothesis Testing Framework

- We start with a null hypothesis (H_0) that represents the status quo.
- We also have an alternative hypothesis (H_A) that represents our research question, i.e. what we're testing for.
- We then collect data and often summarize the data as a test statistic, which is usually a measure gauging whether H_0 or H_A are more plausible
- We then predict what the test statistic would be around under the assumption that the H_0 is true.
- If the test statistic is too far away from what the H_0 predicts, we then reject the H_0 in favor of the H_A .
  - We often compute a p-value based on the test statistic, which is the probability to obtain a test statistic at least as extreme as the one actually observed, assuming the H_0 is true
  - If the p-value is too small, we then reject the H_0 in favor of the H_A .

Case study: student performance

- This dataset captures student achievement in secondary education of two Portuguese schools, specifically in the subject of Portuguese language.
- The dataset attributes include student grades, demographic, social and school related features) and it was collected using school reports and questionnaires.
- Important note: the target attribute G3 has a strong correlation with attributes G2 and G1. This occurs because G3 is the final year grade (issued at the 3rd period), while G1 and G2 correspond to the 1st and 2nd-period grades.

Attribute	Description
school	Student's school (binary: 'GP' - Gabriel Pereira or 'MS' - Mousinho da Silveira)
sex	Student's sex (binary: 'F' - female or 'M' - male)
age	Student's age (numeric: from 15 to 22)
address	Student's home address type (binary: 'U' - urban or 'R' - rural)
famsize	Family size (binary: 'LE3' - less or equal to 3 or 'GT3' - greater than 3)
Pstatus	Parent's cohabitation status (binary: 'T' - living together or 'A' - apart)
Medu	Mother's education (numeric: 0 - none, 1 - primary education (4th grade), 2 – 5th to 9th grade, 3 – secondary education or 4 – higher education)
Fedu	Father's education (numeric: 0 - none, 1 - primary education (4th grade), 2 – 5th to 9th grade, 3 – secondary education or 4 – higher education)
Mjob	Mother's job (nominal: 'teacher', 'health' care related, civil 'services' (e.g. administrative or police), 'at_home' or 'other')
Fjob	Father's job (nominal: 'teacher', 'health' care related, civil 'services' (e.g. administrative or police), 'at_home' or 'other')
reason	Reason to choose this school (nominal: close to 'home', school 'reputation', 'course' preference or 'other')
guardian	Student's guardian (nominal: 'mother', 'father' or 'other')
traveltime	Home to school travel time (numeric: 1 - <15 min., 2 - 15 to 30 min., 3 - 30 min. to 1 hour, or 4 - >1 hour)
studytime	Weekly study time (numeric: 1 - <2 hours, 2 - 2 to 5 hours, 3 - 5 to 10 hours, or 4 - >10 hours)
failures	Number of past class failures (numeric: n if 1<=n<3, else 4)

Attribute	Description
schoolsup	Extra educational support (binary: yes or no)
famsup	Family educational support (binary: yes or no)
paid	Extra paid classes within the course subject (Math or Portuguese) (binary: yes or no)
activities	Extra-curricular activities (binary: yes or no)
nursery	Attended nursery school (binary: yes or no)
higher	Wants to take higher education (binary: yes or no)
internet	Internet access at home (binary: yes or no)
romantic	With a romantic relationship (binary: yes or no)
famrel	Quality of family relationships (numeric: from 1 - very bad to 5 - excellent)
freetime	Free time after school (numeric: from 1 - very low to 5 - very high)
goout	Going out with friends (numeric: from 1 - very low to 5 - very high)
Dalc	Workday alcohol consumption (numeric: from 1 - very low to 5 - very high)
Walc	Weekend alcohol consumption (numeric: from 1 - very low to 5 - very high)
health	Current health status (numeric: from 1 - very bad to 5 - very good)
absences	Number of school absences (numeric: from 0 to 93)
G1	First period grade (numeric: from 0 to 20)
G2	Second period grade (numeric: from 0 to 20)
G3	Final grade (numeric: from 0 to 20, output target)